`GeoModels` Tutorial: simulation, estimation and prediction of spatial data using Gaussian random fields with a compactly supported flexible covariance function

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Introduction

In this tutorial we show how to analyze spatial data using Gaussian random fields with a flexible covariance model proposed in Bevilacqua et al. [2022] using the `R` package `GeoModels` [Bevilacqua et al., 2026].

We first load the R libraries needed for the analysis using the `GeoModels` package:

```
# Install once, if necessary: install.packages("GeoModels")
require(GeoModels)
require(fields)
require(spam)
set.seed(89)
```

Simulation of a spatial Gaussian random field with a flexible compactly supported correlation model

Let us consider a spatial Gaussian random field $Z = \{Z(\mathbf{s}), \mathbf{s} \in S\}$, where $\mathbf{s}$ represents a location site in the domain $S \subset \mathbb{R}^d$ (in this tutorial we consider the case $d = 2$). We assume that Z is stationary with zero mean, unit variance and correlation function given by $\rho(\mathbf{h}) = \text{cor}(Z(\mathbf{s}+\mathbf{h}), Z(\mathbf{s}))$.

Then we consider a random field $Y = \{Y(\mathbf{s}), \mathbf{s} \in S\}$ defined by the location and scale transformation:

$$Y(\mathbf{s}) = m(\mathbf{s}) + \sigma Z(\mathbf{s}). \tag{1}$$

Here the spatial mean in the package `GeoModels` can be specified through a regression model $m(\mathbf{s}) = X(\mathbf{s})^T\boldsymbol{\beta}$. Here $X(\mathbf{s})$ is a k-dimensional vector of covariates and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_k)^T$ is a vector of unknown parameters. Then $\text{E}\{Y(\mathbf{s})\} = X(\mathbf{s})^T\boldsymbol{\beta}$ and $\text{var}\{Y(\mathbf{s})\} = \sigma^2$. With nugget parameter $0 \leq \tau^2 < 1$, the covariance is

$$\text{cov}\{Y(\mathbf{s}+\mathbf{h}), Y(\mathbf{s})\} = \sigma^2 \left[(1-\tau^2)\rho(\mathbf{h}) + \tau^2 \mathbf{1}_{\{\mathbf{h}=\mathbf{0}\}}\right].$$

In this tutorial, for simplicity, we assume a constant spatial mean, unit variance and zero nugget: $m(\mathbf{s}) = m$, $\sigma^2 = 1$ and $\tau^2 = 0$. Additionally we assume that $\rho(\mathbf{h})$ is isotropic, that is, the correlation depends on the Euclidean distance $\|\mathbf{h}\|$; hereafter we set $r = \|\mathbf{h}\|$.

The globally supported Matérn family of covariance functions [Stein, 1999] has played a central role in spatial statistics for decades, being a flexible parametric class with one

parameter determining the smoothness of the paths of the underlying spatial field. It is defined as follows:

$$M_{\nu,\beta}(r) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\frac{r}{\beta}\right)^{\nu} K_{\nu}\left(\frac{r}{\beta}\right), \qquad r \geq 0,$$

for $\nu > 0$, $\beta > 0$, and it is positive definite in any dimension $d = 1, 2, \ldots$. Here Γ is the gamma function and K_{ν} is the modified Bessel function of the second kind of order ν [Abramowitz and Stegun, 1970]. The correlation is globally supported because $M_{\nu,\beta}(r) > 0$ for $r > 0$. The parameter ν indexes the mean-squared differentiability of the Gaussian random field and its associated sample paths. In particular, for a positive integer k, the sample paths are k times differentiable, in any direction, if and only if $\nu > k$ [Stein, 1999, Banerjee et al., 2004].

We now introduce a generalization of the Matérn model proposed in Bevilacqua et al. [2022]. Let

$$\delta_{\nu,\mu,\beta} = \beta \left(\frac{\Gamma(\mu + 2\nu + 1)}{\Gamma(\mu)}\right)^{1/(1+2\nu)}, \tag{2}$$

where $\nu \geq 0$, $\beta > 0$ and $\mu \geq (d+1)/2 + \nu$. Let ${}_2F_1$ denote the Gauss hypergeometric function defined as

$${}_2F_1(a, b; c; x) = \sum_{k=0}^{\infty} \frac{(a)_k (b)_k}{(c)_k} \frac{x^k}{k!}. \tag{3}$$

The correlation model proposed in Bevilacqua et al. [2022] is a specific compact support reparameterization of the generalized Wendland model [Bevilacqua et al., 2019, Gneiting, 2002]. It has a representation in terms of the Gauss hypergeometric function:

$$\mathrm{GW}_{\nu,\mu,\delta_{\nu,\mu,\beta}}(r) = \begin{cases} K\left(1 - \left(\frac{r}{\delta_{\nu,\mu,\beta}}\right)^2\right)^{\nu+\mu} {}_2F_1\left(\frac{\mu}{2}, \frac{\mu+1}{2}; \nu+\mu+1; 1 - \left(\frac{r}{\delta_{\nu,\mu,\beta}}\right)^2\right), & 0 \leq r \leq \delta_{\nu,\mu,\beta}, \\ 0, & r > \delta_{\nu,\mu,\beta}, \end{cases} \tag{4}$$

where $K = \dfrac{\Gamma(\nu)\Gamma(2\nu + \mu + 1)}{\Gamma(2\nu)\Gamma(\nu + \mu + 1)2^{\mu+1}}$. At $\nu = 0$, this representation is understood by continuous extension; the simpler expression is shown in Table 1.

Similarly to the Matérn case, this model allows for a continuous parameterization of the mean-squared differentiability of the underlying Gaussian random field and its associated sample paths. Specifically, the sample paths of the generalized Wendland model are k times differentiable, in any direction, if and only if $\nu > k - 0.5$.

The correlation model (4) is very flexible, as it allows us to consider, under the same umbrella, compactly and globally supported correlation functions. In fact, Bevilacqua et al.

[2022] show that the Matérn family $M_{\nu+1/2,\beta}$ is obtained as the limit of the $\mathrm{GW}_{\nu,\mu,\delta_{\nu,\mu,\beta}}$ model when $\mu \to \infty$. Hence μ controls the sparsity of the associated correlation matrix and allows the transition from flexible compactly supported covariance functions to globally supported ones. Note that $\delta_{\nu,\mu,\beta}$ depends on ν, β and μ and is an increasing function of μ.

In Table 1 we report the correlation models $\mathrm{GW}_{\nu,\mu,\delta_{\nu,\mu,\beta}}$ for $\nu = 0, 1, 2, 3$ and the associated limit cases as $\mu \to \infty$, that is, the Matérn models $M_{\nu+1/2,\beta}$.

Suppose we want to simulate a realization of a random field Y with constant mean equal to five, unit variance and correlation model (4) at $N = 500$ spatial locations uniformly distributed in the unit square. We first set the spatial coordinates:

```
NN=500                # number of spatial locations
x=runif(NN,0,1); y=runif(NN,0,1)
coords=cbind(x,y)
X=cbind(rep(1,NN),runif(NN)) # unused; preserves original R seed sequence
```

Then we specify the mean, variance and nugget parameters for the Gaussian random field:

```
GeoModels::NuisParam("Gaussian")
mean=5; sill=1; nugget=0
```

where `mean`, `sill` and `nugget` correspond to m, σ^2 and τ^2, respectively. Then we set the name of the correlation model and its parameters:

```
corrmodel="GenWend_Matern"
GeoModels::CorrParam(corrmodel)
mu=3
scale=0.05; power2=1/mu; smooth=0
```

where `scale`, `power2` and `smooth` correspond to β, $1/\mu$ and ν, respectively. The compact support for $\nu = 0$ is $\delta_{0,\mu,\beta} = \mu\beta$, calculated as

```
scale*(1/power2)
```

The result is 0.15. We are now ready to simulate the spatial Gaussian random field using the function `GeoSim`:

```
param=list(nugget=nugget,mean=mean,scale=scale,
           sill=sill,power2=power2,smooth=smooth)
sim=GeoModels::GeoSim(coordx=coords,corrmodel=corrmodel,
      sparse=FALSE,model="Gaussian",param=param)$data
```

ν	$\mathcal{GW}_{\nu,\mu,\delta_{\nu,\mu,\beta}}(r)$	$\mathcal{M}_{\nu+1/2,\beta}(r)$
0	$\left(1-\frac{r}{\delta_{0,\mu,\beta}}\right)_+^{\mu}$	$e^{-\frac{r}{\beta}}$
1	$\left(1-\frac{r}{\delta_{1,\mu,\beta}}\right)_+^{\mu+1}\left(1+\frac{r}{\delta_{1,\mu,\beta}}(\mu+1)\right)$	$e^{-\frac{r}{\beta}}(1+\frac{r}{\beta})$
2	$\left(1-\frac{r}{\delta_{2,\mu,\beta}}\right)_+^{\mu+2}\left(1+\frac{r}{\delta_{2,\mu,\beta}}(\mu+2)+\left(\frac{r}{\delta_{2,\mu,\beta}}\right)^2(\mu^2+4\mu+3)\frac{1}{3}\right)$	$e^{-\frac{r}{\beta}}(1+\frac{r}{\beta}+\frac{r^2}{3\beta^2})$
3	$\left(1-\frac{r}{\delta_{3,\mu,\beta}}\right)_+^{\mu+3}(1+\frac{r}{\delta_{3,\mu,\beta}}(\mu+3)+\left(\frac{r}{\delta_{3,\mu,\beta}}\right)^2(2\mu^2+12\mu+15)\frac{1}{5}$ $+\left(\frac{r}{\delta_{3,\mu,\beta}}\right)^3(\mu^3+9\mu^2+23\mu+15)\frac{1}{15})$	$e^{-\frac{r}{\beta}}(1+\frac{r}{2\beta}+\frac{6r^2}{15\beta^2}+\frac{r^3}{15\beta^3})$

Table 1: The $\mathrm{GW}_{\nu,\mu,\delta_{\nu,\mu,\beta}}$ model with compact support $\delta_{\nu,\mu,\beta}$ (Equation 2) for $\nu = 0, 1, 2, 3$ and its limiting Matérn model $M_{\nu+1/2,\beta}$.

Note that the option `sparse` controls whether sparse matrix algorithms are used for Cholesky decomposition, through the `spam` package [Gerber et al., 2017]. Information about the sparsity of the covariance matrix can be obtained using `GeoCovmatrix`:

```
cc=GeoModels::GeoCovmatrix(coordx=coords,corrmodel=corrmodel,
       sparse=TRUE,model="Gaussian",param=param)
is.spam(cc$covmatrix)
cc$nozero
```

In the original example, the proportion of nonzero entries was 0.062168. Thus, approximately 94% of the covariance matrix entries were zero. To visualize the sparsity pattern we use

```
GeoCovDisplay(cc)
```

Here white indicates zero entries of the covariance matrix.

Maximum likelihood estimation

Given a realization $\mathbf{Y} = (Y(\mathbf{s}_1), \ldots, Y(\mathbf{s}_N))^T$, maximum likelihood estimation is obtained by maximizing, up to an additive constant,

$$\ell(\boldsymbol{\theta}) = -\frac{1}{2}\log|\sigma^2\mathbf{R}| - \frac{1}{2\sigma^2}(\mathbf{Y} - m\mathbf{1})^T\mathbf{R}^{-1}(\mathbf{Y} - m\mathbf{1}),$$

where $\mathbf{R} = [\mathrm{GW}_{\nu,\mu,\delta_{\nu,\mu,\beta}}(\|\mathbf{s}_i - \mathbf{s}_j\|)]_{i,j=1}^N$. For this example, $\nu = 0$ and the nugget parameter are fixed, and the estimated parameter vector is $\boldsymbol{\theta} = (m, \sigma^2, \beta, 1/\mu)^T$. Estimation can

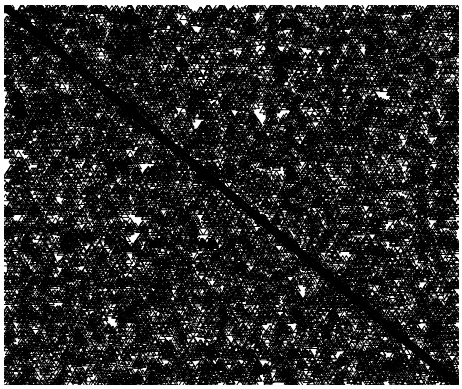


Figure 1: Pattern of the sparsity of the covariance matrix (white indicates zero entries).

be performed with `GeoFit`. We use the optimization method `nlminb`, which allows box constraints.

```
optimizer="nlminb"
start=list(mean=mean,scale=scale,sill=sill,power2=power2)
I=Inf
lower=list(mean=-I,scale=1e-5,sill=1e-5,power2=1e-4)
upper=list(mean=I,scale=I,sill=I,power2=1/1.5)
fixed=list(nugget=nugget,smooth=smooth)
fitML <- GeoModels::GeoFit(data=sim,coordx=coords,
    corrmodel=corrmodel,model="Gaussian",varest=TRUE,
    optimizer=optimizer,lower=lower,upper=upper,
    likelihood="Full",type="Standard",start=start,
    fixed=fixed)
```

The object `fitML` includes information about the maximum likelihood estimates. In the original PDF, the following output was reported (numerical results may differ across package versions or random number streams):

```
fitML
# Maximum Likelihood Fitting of Gaussian Random Fields
# Setting: Full Likelihood
# Model: Gaussian
# Covariance model: GenWend_Matern
```

```
# Optimizer: nlminb
# Number of spatial coordinates: 500
# Number of estimated parameters: 4
# Type of convergence: Successful
# Maximum log-Likelihood value: -593.19
# AIC: 1194    BIC: 1211
# Estimated parameters:
#     mean     power2     scale      sill
#   5.04396    0.43166   0.04587   0.96836
# Standard errors:
#     mean     power2     scale      sill
#   0.075632   0.103227  0.004547  0.078177
```

In the original example the estimated compact support was:

```
fitML$param[["scale"]] / fitML$param[["power2"]]
```

which was reported as 0.1062629.

Checking model assumptions

Given the estimated mean and sill, the standardized residuals

$$\widehat{Z}(\mathbf{s}_i) = \frac{Y(\mathbf{s}_i) - \widehat{m}}{\sqrt{\widehat{\sigma^2}}}, \qquad i = 1, \ldots, N,$$

can be viewed as one realization of a standard Gaussian random field with correlation function (4). The residuals can be computed using `GeoResiduals`:

```
res=GeoModels::GeoResiduals(fitML)
```

Then the marginal distribution assumption on the residuals can be checked graphically using a QQ-plot (Figure 2, left):

```
GeoQQ(res)
```

The correlation model assumption can be checked by comparing the empirical and estimated semivariograms (Figure 2, right) using `GeoVariogram` and `GeoCovariogram`:

```
vario=GeoModels::GeoVariogram(data=res$data,
                              coordx=coords,maxdist=0.6)
GeoCovariogram(res,vario=vario,
                           show.vario=TRUE,pch=20)
```

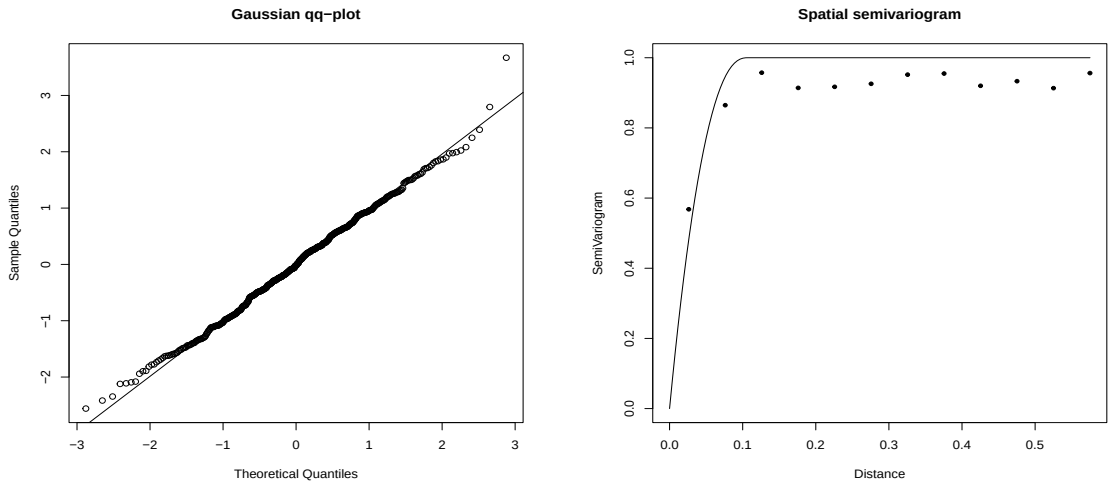


Figure 2: Left: QQ-plot for the Gaussian random field residuals. Right: empirical versus estimated semivariogram.

Prediction

For a given spatial location $\mathbf{s}_0$, the optimal linear predictor, using estimated parameters, is computed as

$$\widehat{Y}(\mathbf{s}_0) = \widehat{m} + \mathbf{c}^T\mathbf{R}^{-1}[\mathbf{Y} - \widehat{m}\mathbf{1}], \tag{5}$$

where $\mathbf{c} = (\text{cor}\{Y(\mathbf{s}_0), Y(\mathbf{s}_1)\}, \ldots, \text{cor}\{Y(\mathbf{s}_0), Y(\mathbf{s}_N)\})^T$ and $\mathbf{R} = [\text{cor}\{Y(\mathbf{s}_i), Y(\mathbf{s}_j)\}]_{i,j=1}^N$. Kriging can be performed using `GeoKrig`. In this example, we consider a regular spatial prediction grid:

```
xx=seq(0,1,0.015)
loc_to_pred=as.matrix(expand.grid(xx,xx))
```

The optimal linear prediction (5) is computed as

```
pr=GeoModels::GeoKrig(fitML,loc=loc_to_pred,
                      sparse=TRUE,mse=TRUE)
```

A kriging map and the associated mean squared prediction error are shown in Figure 3 and can be obtained with:

```
quilt.plot(coords[,1],coords[,2],sim,main="Observed")
image.plot(xx,xx,matrix(pr$pred,ncol=length(xx)),
                   main="Prediction",ylab="")
image.plot(xx,xx,matrix(pr$mse,ncol=length(xx)),
```

```
                    main="MSE",ylab="")
```

Here `GeoKrig` performs simple kriging using fitted parameters. Its reported MSE is a plug-in prediction MSE, not a measure incorporating all covariance-parameter estimation uncertainty.

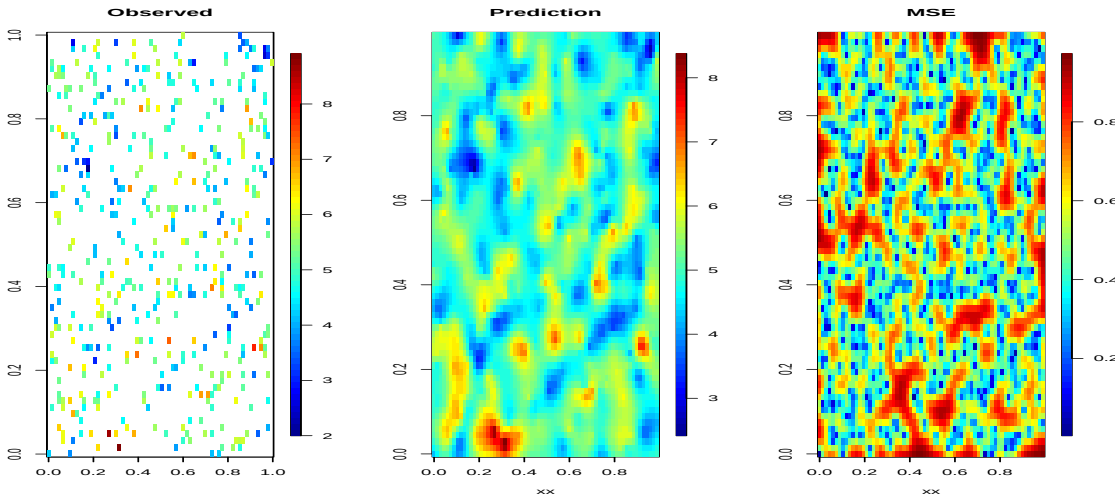


Figure 3: From left to right: observed spatial data, estimated kriging map and associated mean squared prediction error map.

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